SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I U. Ananya (192626382) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: U. Ananya.

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 24 **/30**

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

CO1 - QT2

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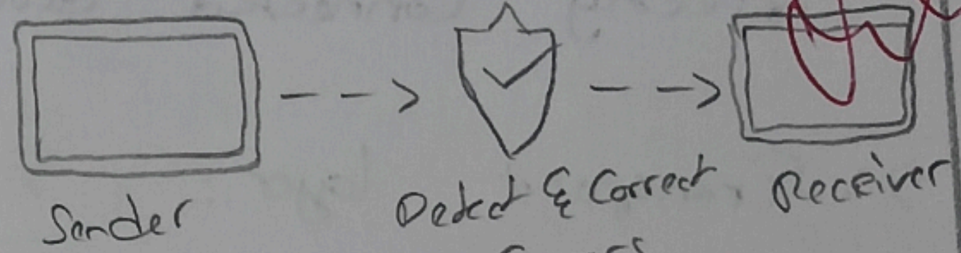
Error Control

- Improves reliability in data transmission -

what is it?

Error control is a mechanism used in communication networks to detect and correct errors in data during transmission, ensuring reliable and accurate delivery.



How it works (layer by layer)

2. Data Link layer

Error Detection

Detects error using CRC and checksums.

4. Transport layer

Error Recovery and Reliable Delivery

uses acknowledgements and retransmissions in TCP to recovery lost data.

Reliable data Transmission.

Correct and complete data reaches the destination.

Analysis:

1- Data link layer:

Detects errors in frames during transmissions between directly connected devices.

2. Transport layer:

TCP helps ensure reliable delivery by detecting missing data and retransmitting it when necessary.

3. Interaction:

The Data link layer handles local transmission errors, while the Transport layer provides end-to-end reliability.

4. Result:

Both layers work together to improve data accuracy and reliablity.

key Takeaways:

* Error control uses detections and corrections machanisms.

* Data link layer finds error.

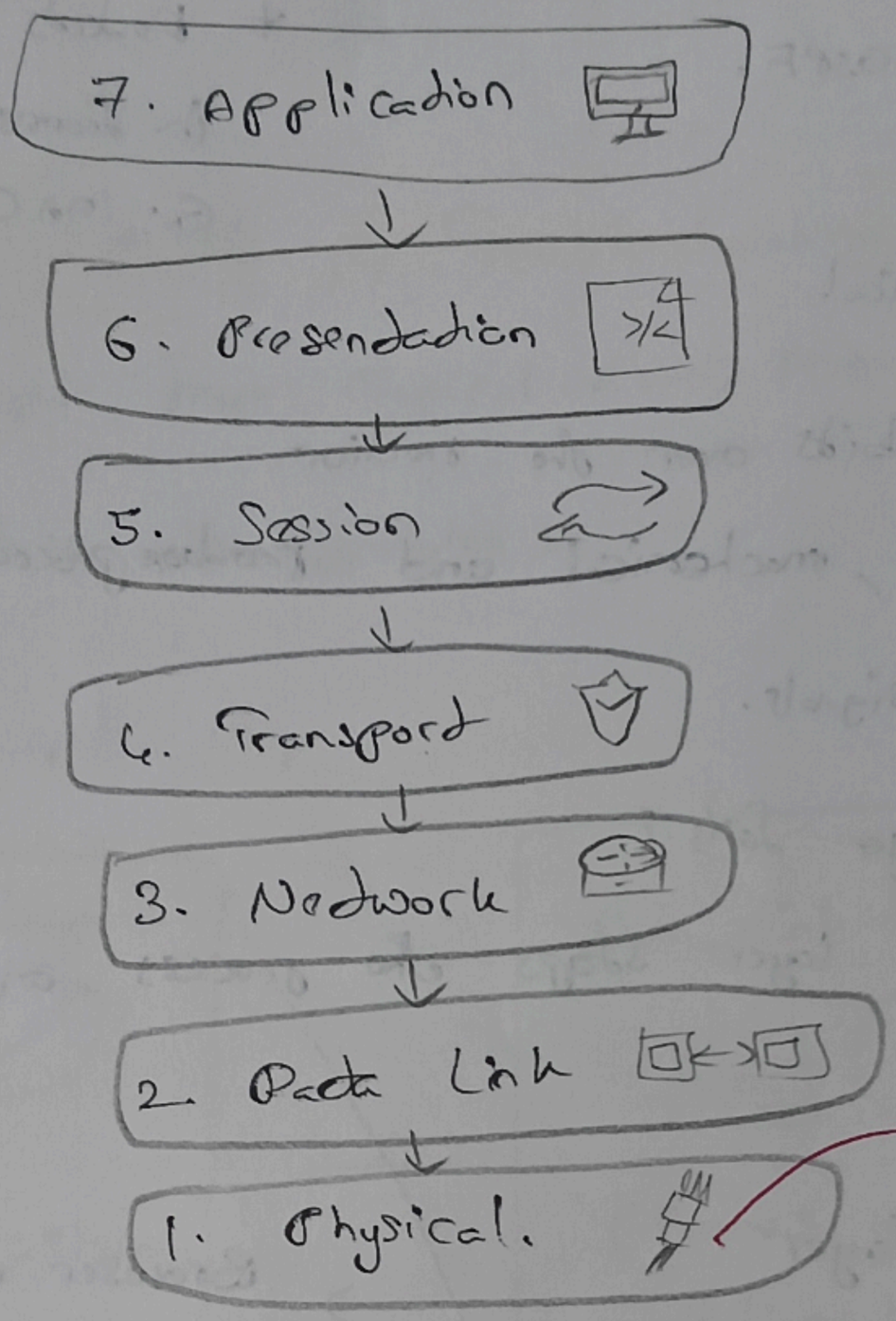
* Transport layer recovers from errors

* Together they ensure reliable and accurate data delivery.

OSI Model

(Open systems Interconnection)

7 Layers → Reliable Communication.



→ layer 7 : Application layer

* Provides network services to end-user applications.

Ex: HTTP, FTP, SMTP, DNS

→ layer 6 : Presentation layer

* Handles data formats, translations, and encryption/decryption

* Ex : JPEG, ASCII.

→ layer 5 : Session layer

* Manages session establishment maintenance and terminates

* Provides synchronization and recovery

Ex: NetBIOS, RPC.

→ layer 4 : Transport

* Provides end-to-end delivery

* Handles error detections and correction

* Ex: TCP, UDP.

⟶ layer 3: Network layer

* Handles logical addressing and routing
* Determines the best path of data
* Ex: IP, ICMP, OSPF.

⟶ layer 2: Data link

* Provides reliable delivery over the same network segment
* Detects and corrects errors in frames
  Ex: MAC, PPP, Ethernet..

⟶ layer 1: Physical

* Transmits raw bits over the medium
* Defines electrical, mechanical and procedural structural
* Ex: Cables, hubs, signals.

What if one layer fails?

A failure in any layer stops the process and effects all the layers above it.

7. Application layer (e.g: DNS or HTTP error) ⟶ Browser can't find the website or load the page
6. Presentation layer (e.g., SSL/TLS error) ⟶ Data can't be read or verified
5. Session layer (e.g., session timeout) ⟶ Connection gets dropped, no data exchanged
4. Transport layer (e.g. TCP port blocked) ⟶ Data is not delivered reliability.
3. Network layer (e.g., wrong IP, no route) ⟶ Packet can't reach the destination

2. Data link layer (e.g., wrong MAC) → Devices can't connect to the local network

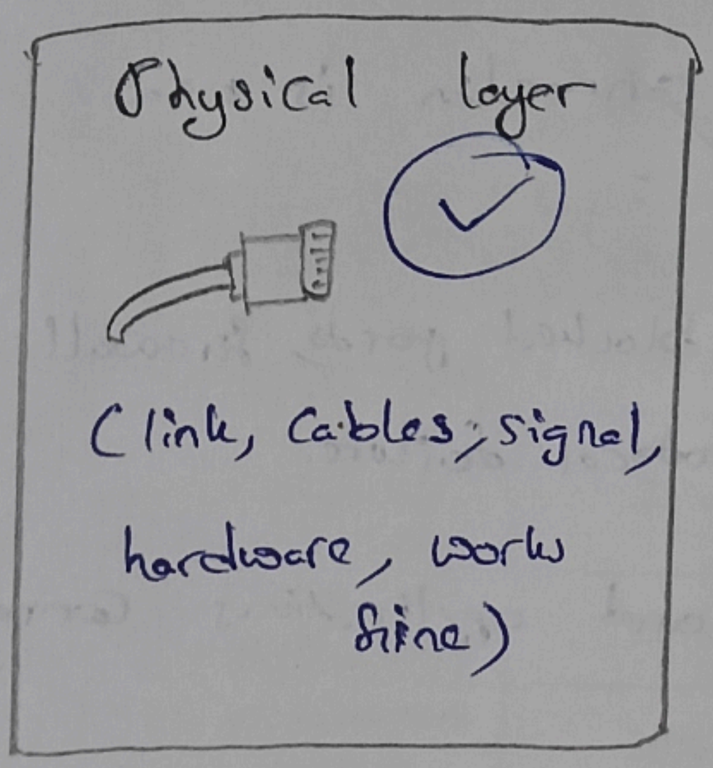
1. Physical layer (e.g., cable unplugged) → No signal or data transmission at all.

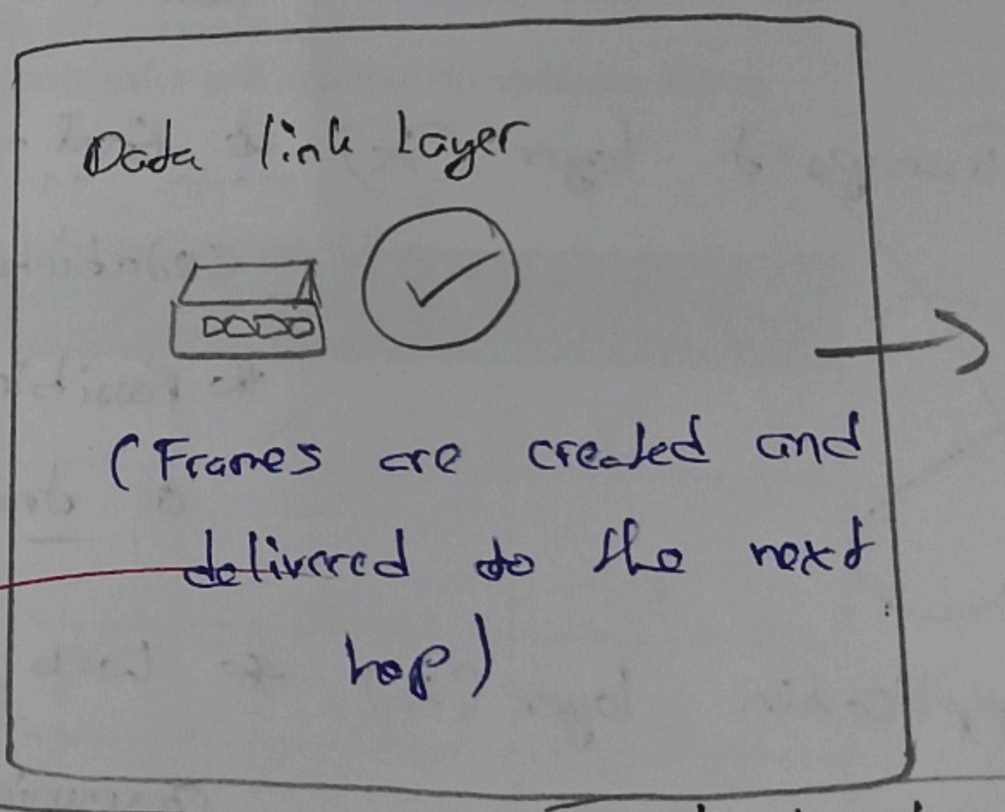
6.

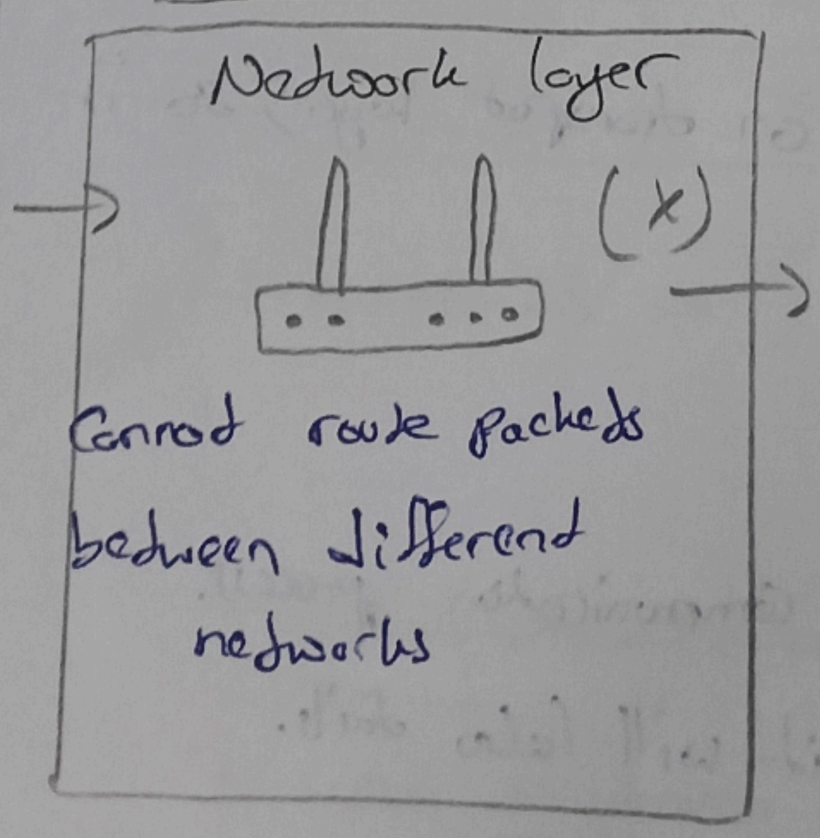
Network Issue Analysis & Troubleshooting.

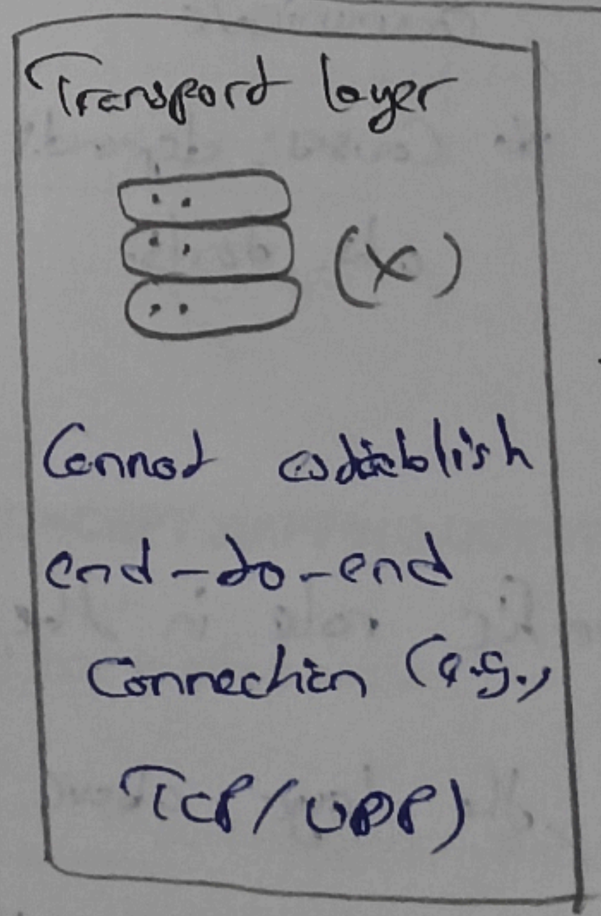
Using the Concept map: Physical ✓ → Data link ✓ → Network ✗ → Transport ✗ → Application ✗

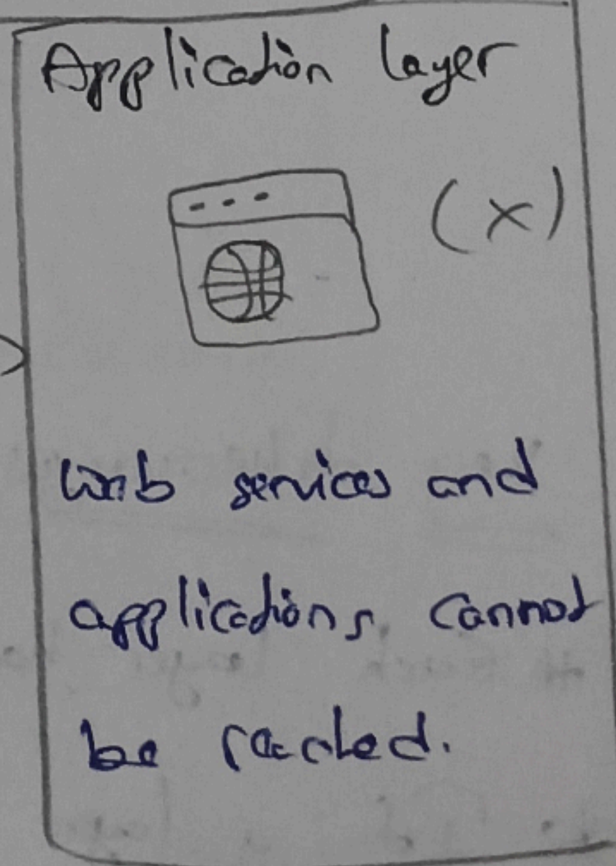
The given Concept map











What the problem means.

Physical layer (✓) * The physical connection is fine.
* Devices can send and receive electrical / optical signals.

Data link layer (✓) * Frames are being created and delivered to the next devices.

Network layer (✗) * Packet cannot be routed to other networks
* Possible causes: wrong IP/gateway, routing issue or networking devices failure

Transport layer (✗) * End-to-end connection is not established
* Possible causes: blocked ports, firewall rules, or transport protocol failure.

Application layer (✗) * Web services and applications cannot communicate
* Causes: depends on transport layer, so it also fails.

Key takeaways:

* Each layer has a specific role in the communication process.
* If a layer fails, the layer above it will also fails.
* The concept map helps identify the exact layer causing the issue.